

3. **Cost management:** The most difficult and time consuming aspect to understand in multi-cloud services is cost management. Cloud providers charge end users on the basis of computing, storage, networking, region, data transmission and other parameters. Some cloud providers not only charge for the size of the load balancer but also for the type of instance – spot or reserved, how many requests an instance will handle per server per second, whether the load balancer will move data from one region to another, and how many rules a load balancer can apply to move the data. Charges for all of these aspects vary from cloud to cloud, and definitions of services do not even match up perfectly. For example, in GCP, there are different tiers for networking, which do not exist in Azure. Managing and forecasting costs in one cloud is also challenging but often made more addressable by the providers' management and projection tools.
4. **Prolonged time on application changes:** Deployment of new applications on multiple clouds can even slow down features and functionality as we need more time for testing the new changes in operational cloud environments. Sometimes, containerised applications perform differently on different clouds, and for mission-critical applications, we need more time and budget to test the performance.
5. **Security risks and attacks:** In multi-cloud environments, security teams need to monitor three times more services in the cloud, which sometimes leads to mistakes and opens gates for hackers to attack. Security teams will also need to configure and test 2x or 3x more security appliances and tools. This creates a lot more stress on already overloaded security teams, and increases the likelihood of human error resulting from a misconfiguration or a missed update. DevOps teams dealing with multiple clouds may get frustrated with the complexity, and create workarounds that increase the attack surface area and add risk. In addition, data moving between clouds means more exposure and an increased attack surface.

With a multi-cloud strategy, you can benefit from the best that each cloud has to offer. But this comes with its own set of challenges and considerations. That's why it is desirable to evaluate and identify the right set of management tools, especially for security and compliance.

Open source multi-cloud management tools

The following are the best open source multi-cloud management tools.

Mist

Mist attempts to make multi-cloud management simple and offers a single interface from where you can manage everything. It supports all popular infrastructure technologies including public clouds, private clouds, hypervisors, containers and bare metal servers. It provides a unified

interface for performing common management tasks like provisioning, orchestration, monitoring, automation and cost analysis. It comes with a RESTful API and CLI so you can easily integrate it into your existing workflows.

Features:

- Instant visibility of all the available resources across clouds, grouped by tags
- Instant reporting/estimation of the current infrastructure costs
- Compare current and past costs, correlate with usage, provide right-sizing recommendations (EE/HS only)
- Provision new resources on any cloud: machines, volumes, networks, zones, records
- Perform life cycle actions on existing resources: stop, start, reboot, resize, destroy, etc
- Instant audit logging for all actions performed through Mist or detected through continuous polling
- Upload scripts to the library, run them on any machine while enforcing audit logging and centralised control of SSH keys
- SSH command shell on any machine within the browser or through the CLI, enforcing audit logging and centralised control of SSH keys
- Enable monitoring on target machines to display real-time system and custom metrics, and store them for long-term access
- Set rules on metrics or logs that trigger notifications, webhooks, scripts or machine life cycle actions
- Set schedules that trigger scripts or machine life cycle actions
- Set fine grained access control policies per team/tag/resource/action (EE/HS only)
- Set governance constraints; e.g., quotas on cost per user/team, required expiration dates (EE/HS only)
- Upload infrastructure templates that may describe complex deployments and workflows (EE/HS only)
- Deploy and scale Kubernetes clusters on any supported cloud (EE/HS only)

Official website: <https://mist.io/>

Latest version: 4.5.5

Cloudify

Cloudify is a pure-play, standards based (TOSCA) cloud orchestration platform that supports every major private and public cloud infrastructure offering. With Cloudify, enterprises can use a single, open source cloud orchestration platform across OpenStack, VMware or AWS clouds, with virtualisation approaches such as VMs or containers, and with different automation tool sets like Puppet, Chef or SaltStack. Because it provides an easy-to-use, open source tool for management and orchestration (MANO) of multiple clouds, data centres and availability zones, Cloudify is attractive to telecoms, Internet service providers, and enterprises using hybrid cloud.